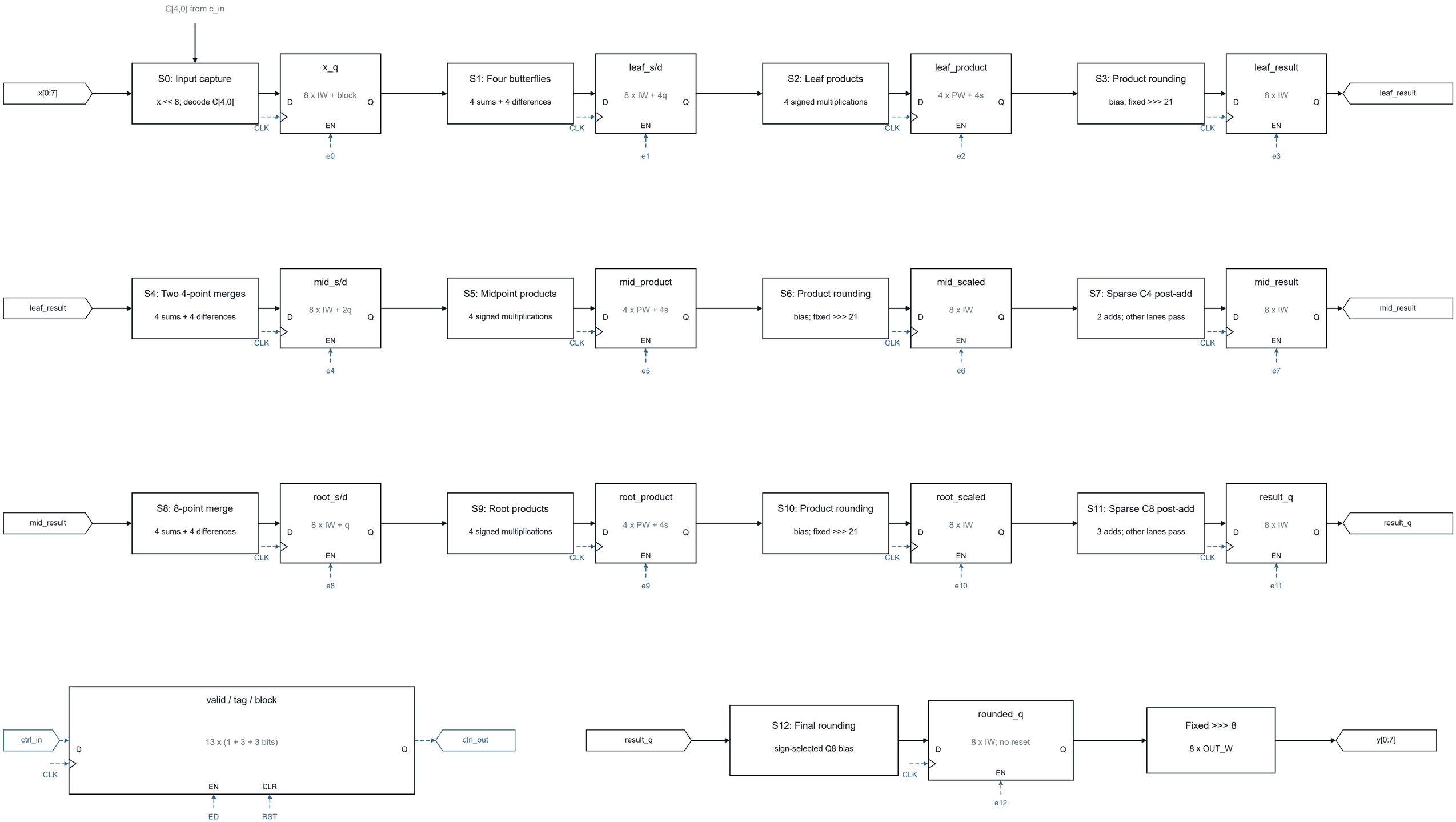
SkewDCT8 core - recursive arithmetic with registered boundaries

USE_COEFF_PORT=2; unchanged ports and original parameters. 12 real products; II=1 enabled edge; 13 enabled register stages.

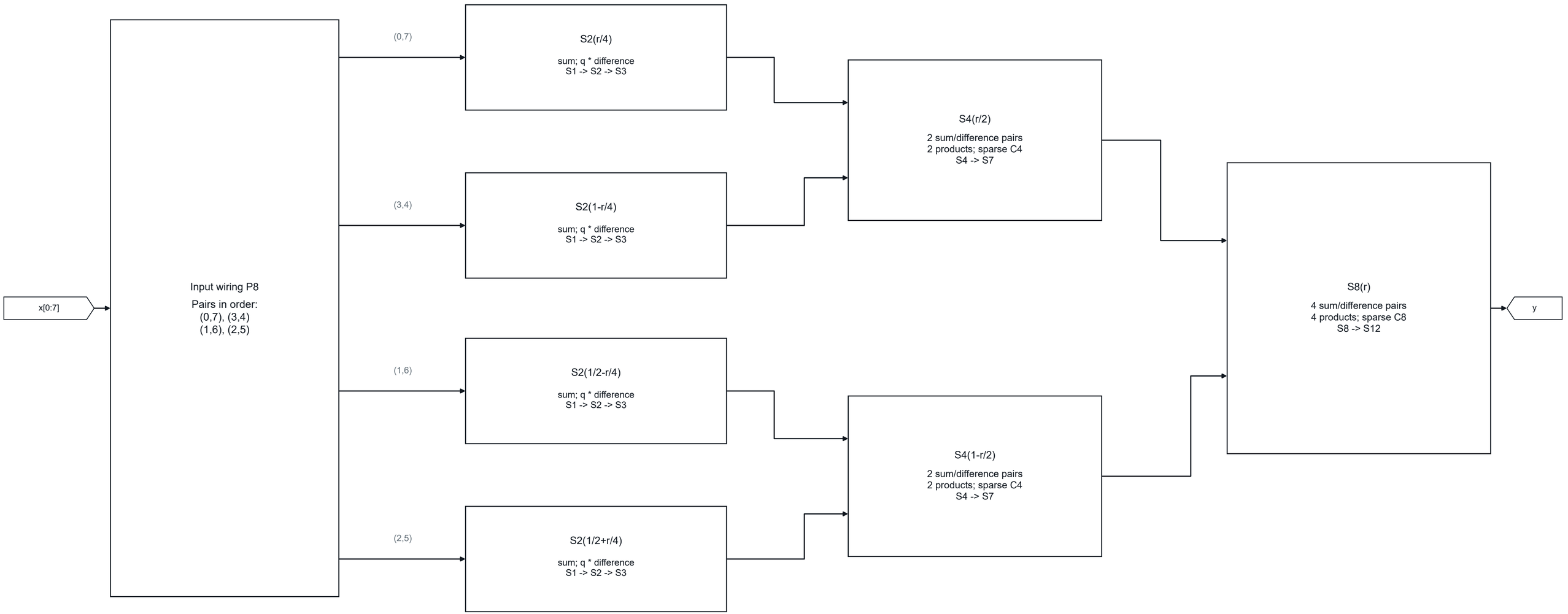


IW=DATA_W+16 (8 fractional guard bits); PW=IW+27; factors are signed Q21. Data registers have no reset; synchronous RST clears control state.

e0 = ~RST & ED & in_valid; ej = ~RST & ED & valid_q[j-1]. ctrl_in = in_valid, in_tag, decoded block; ctrl_out = out_valid, out_tag.

Recursive SkewDCT-II wiring - 8 points to 4 points to 2 points

Conceptual grouping of the registered datapath on R01/R03. r=(2*block+1)/16; q(t)=1/[2*cos(pi*t/2)].

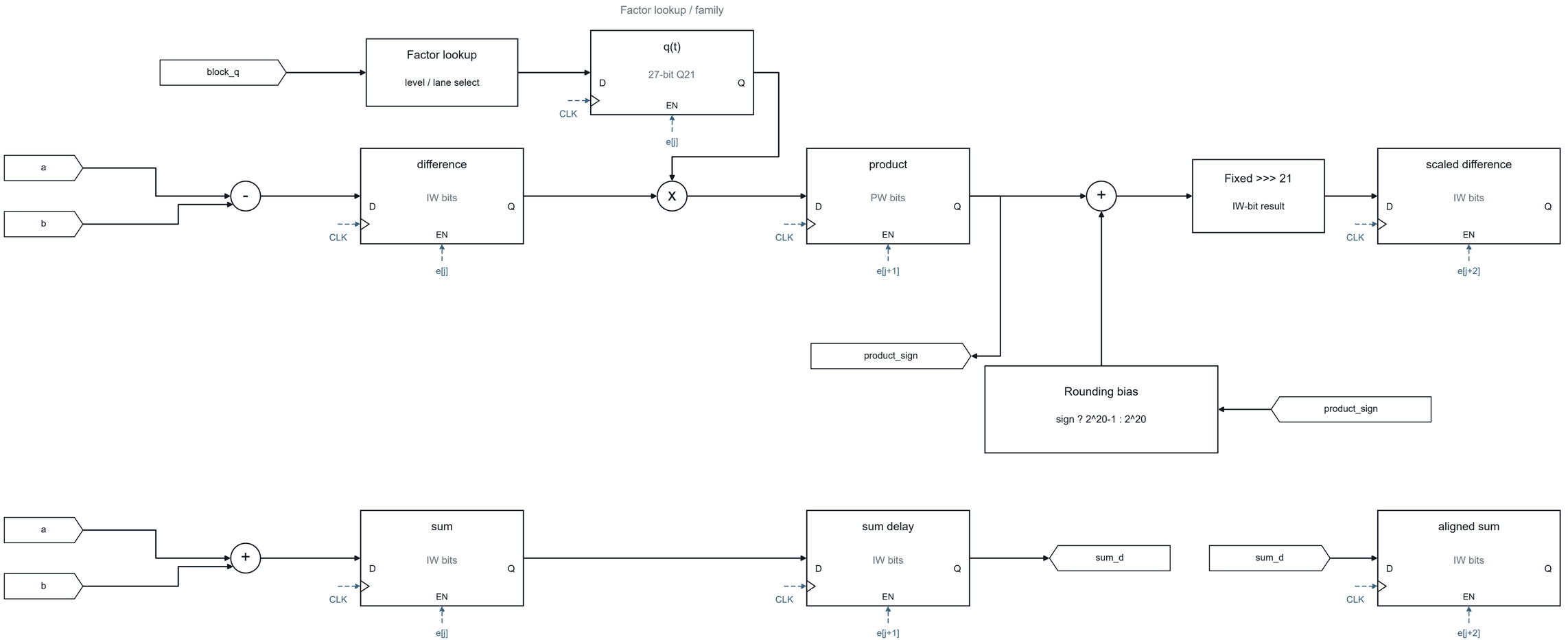


C4: z1 <- z1+z3. C8: z1 <- z1+z7; z2 <- z2+z6; z3 <- z3+z5. All other outputs pass through.

The input permutation is wiring only. Arithmetic count excludes fixed-point rounding: 4+4+4 products and 8+10+11 add/sub operations.

Registered butterfly / product section - no combinational MAC

Four parallel copies per product level. This section repeats at S1-S3, S4-S6 and S8-S10; C4/C8 post-adds have separate registers.

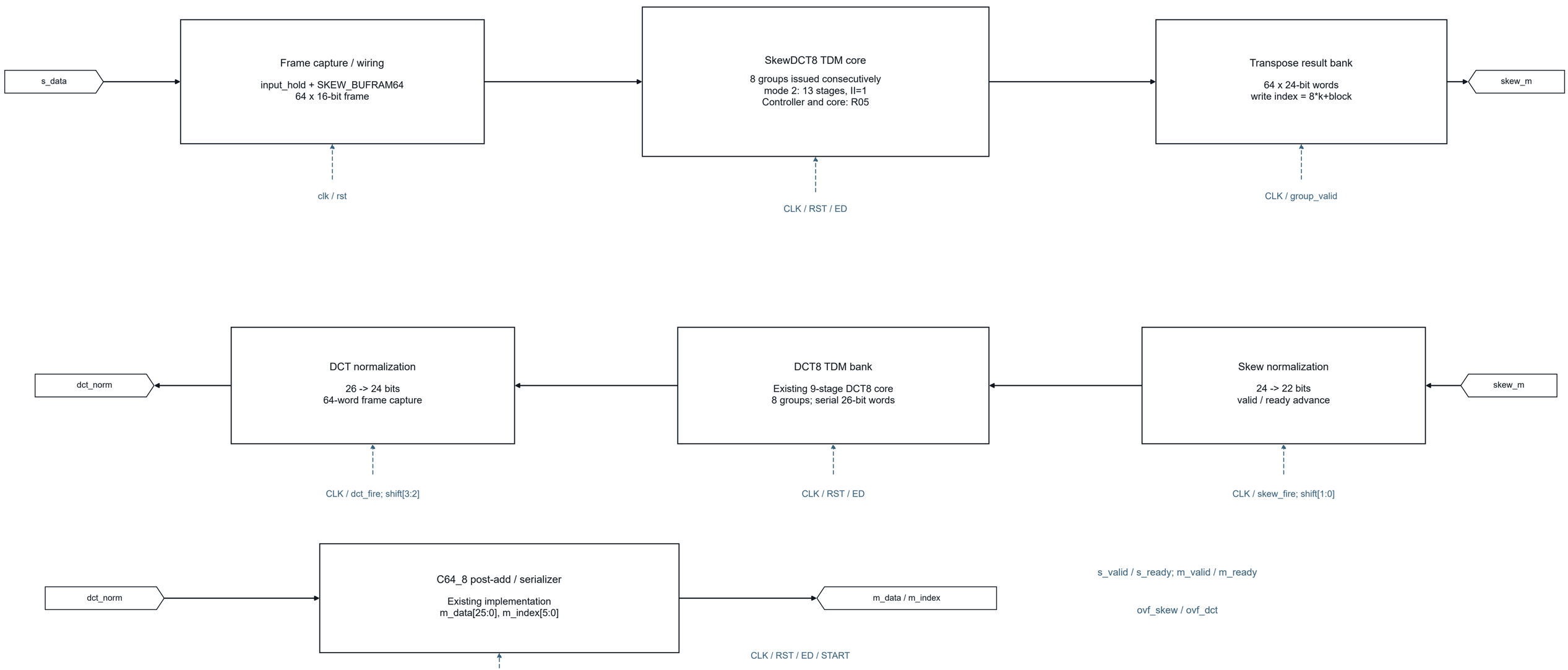


A product is registered before its rounding adder. The sum path has matching delay registers; clock-enable pauses preserve alignment.

j = 1, 4 or 8. Data registers have CLK and EN; reset is applied to the shared valid/tag pipeline shown on R01.

DCT64 integration - recursive Skew bank and existing downstream stages

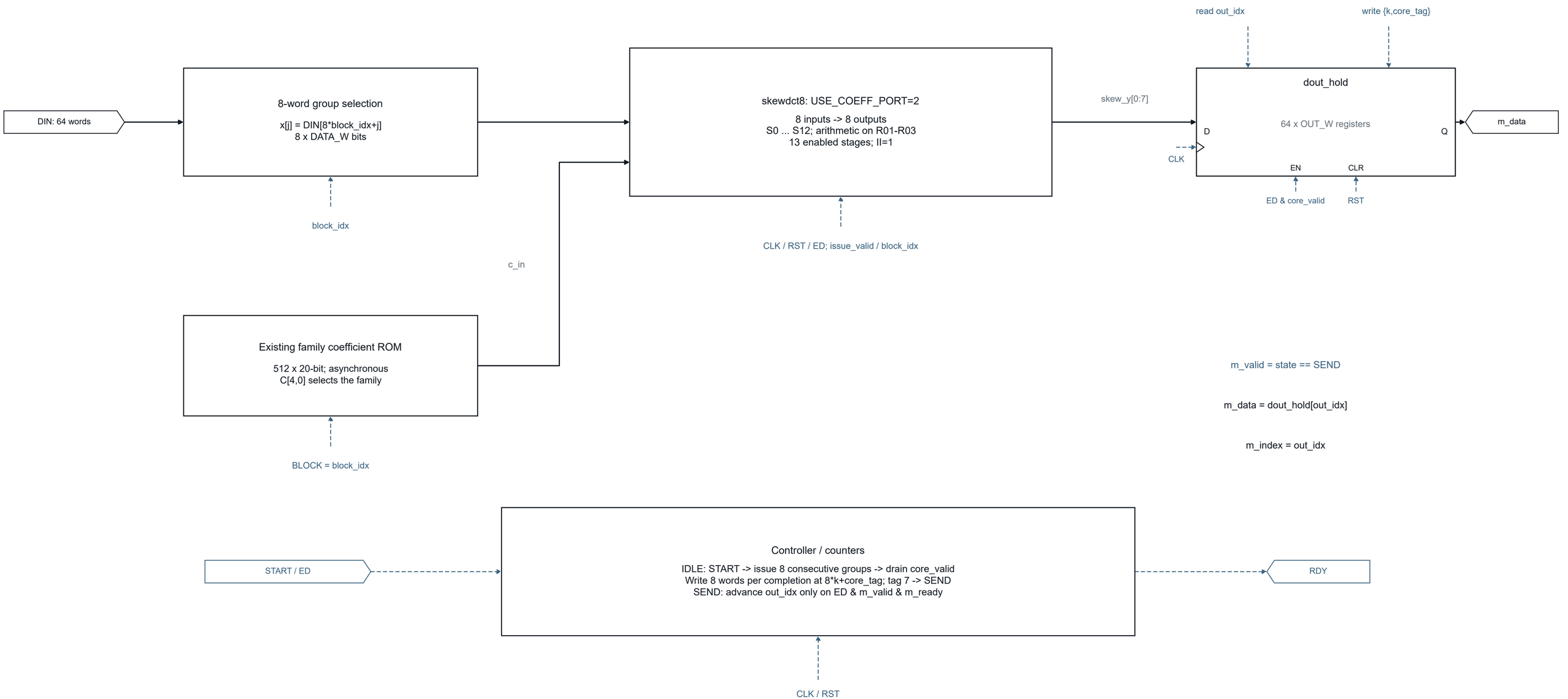
The complete DCT remains in src/work/DCT40_64. Only skewdct_8.v and the bank selection in skewdct8_tdm.v change.



The controller drains out_valid/out_tag, without a fixed core-latency assumption. No-stall DCT64 first output: 165 -> 172 cycles in the unit test.

SkewDCT8 TDM bank - one recursive core shared by eight groups

skewdct8_dctii_tdm_bank -> skewdct8_dctii_dyn -> skewdct8. The packed input frame must remain stable while the bank is busy.



RST clears controller, counters and result registers. ED freezes issue, core state and serializer. in_tag tracks the group; family selection uses C[4,0].